

Definition

cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet. it allows users to access resources on demand without owning physical characteristics.

on-Demand self-service: users can create and use computing resources whenever required without human intervention.

2) Broad network access: cloud services are accessible through the internet using devices such as laptop, smart phones, and tablets.

3) Resource pooling: The cloud provides share computing resources among multiple users using virtualization improving efficiency.

4) Rapid elasticity:

Resources can be increased or decreased automatically according to workload, making the cloud highly scalable.

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1. Mainframe computers (1960s) :-

Many user shared on the a single powerful computing through terminals.

2. Personal computers (1980s) :-

Individual users shared the using their own computer.

3 Client server Architecture

Application were divided between client systems and central servers.

4) Internet Revolution (1990s) :-

The internet enable communication and online services worldwide.

5) Virtualization Technology :-

One physical server could run multiple virtual machines, improving resource virtualization.

3) Role of server virtualization

server virtualization divides one physical server into multiple virtual machine (vms) Each vm works independently with its own operating system and application

Benefits:

- Better hardware utilization
- lower infrastructure cost
- Easier backups and recovery.
- Faster deployment of application.
- support cloud scalability.

Difference:-

Aspect	virtualization	cloud computing
Def	Technology to run OS on hardware	service model to deliver
Scope	can be used on-Prem	Always delivers as a services.
features	only resource abstract	Add self services billing classify.

4 Data center are the Physical infrastructure behind cloud.

1 Resource hosting: contain servers, storage, and networking equipment used to run cloud services.

2 Reliability.

Provide - power backup, cooling and redundancy

3) Global presence:

cloud providers have regions and availability zones for low latency and disaster recovery.

4) Security:

Physical security, firewalls, and compliance certifications are managed by data centers.

5) Connectivity:

High-speed internet and interconnection allow user to access services.

(5) Service model	Description	Example
IaaS (Infrastructure as a Service).	Provides, virtual server storage and networking	Amazon EC2
PaaS (Platform as a Service)	Provides a platform for developing software	Google Docs Microsoft 365
SaaS (Software as a Service)	Provides a platform for developing and deploying through a web browser.	Google APP engine